

A Necessary and Sufficient Condition for Exact Relaxation of Energy Storage Power Dispatch

Shan Liu, *Student Member, IEEE*, Zhigang Li, *Senior Member, IEEE*, and Ye Guo, *Senior Member, IEEE*

Abstract—Mixed-integer programming (MIP) precludes simultaneous charging and discharging in power dispatch of energy storage system (ESS), but is computationally intensive. Relaxation of this complementarity constraint improves computational efficiency, but may incur an impractical control SCD strategy. To bridge this gap, this letter proposes a necessary and sufficient condition for the exactness of such relaxation. We explicitly derive the relaxation gap and reveal that SCD does not necessarily imply a distinguished optimal value from the MIP model. In such cases, a feasible solution without SCD can be readily recovered without compromising the optimal value. Furthermore, a two-stage relaxation-and-recovery algorithm is proposed to improve the computational efficiency of ESS power dispatch optimization.

Index Terms—Energy storage, complementarity constraint, exact relaxation, power dispatch.

I. INTRODUCTION

Complementarity constraints are conventionally included in energy storage system (ESS) models to exclude impractical simultaneous charging and discharging (SCD) by formulating a mixed-integer programming (MIP) problem, which is acknowledged to be computationally expensive [1]. Without introducing binary variables, direct relaxation of the complementarity constraints avoids this dilemma effectively [2], [3]. However, such relaxation is not necessarily exact, i.e., the relaxed model may yield an impractical SCD solution with a more optimistic optimal value than that of the original problem. Moreover, the existing relevant literatures do not theoretically distinguish whether the SCD of a relaxed solution implies a nonzero relaxation gap or only a removable degeneracy. To bridge these gaps, we establish a necessary and sufficient condition for the exact relaxation of ESS power dispatch problems with complementarity constraints. We further analyze the impact of SCD on optimality and propose a two-stage relaxation-and-recovery algorithm to improve computational efficiency.

S. Liu and Z. Li are with the School of Science and Engineering, The Chinese University of Hong Kong (Shenzhen), Shenzhen, China. Y. Guo is with the Department of Building Environment and Energy Engineering, The Hong Kong Polytechnic University, Hong Kong, China. (Correspondence to Zhigang Li. e-mail: lizg@cuhk.edu.cn)

II. PROBLEM FORMULATION AND ANALYSIS

Consider the power dispatch problem of a single ESS over periods $\mathcal{T} := \{1, \dots, T\}$

$$Q := \min \sum_{t \in \mathcal{T}} (C_t^{\text{CH}} p_t^{\text{ch}} + C_t^{\text{DC}} p_t^{\text{dc}}) \quad (1a)$$

$$\text{s.t. } e_t = e_{t-1} + \eta^{\text{ch}} p_t^{\text{ch}} \Delta t - p_t^{\text{dc}} \Delta t / \eta^{\text{dc}}, \quad t \in \mathcal{T}, \quad (1b)$$

$$0 \leq p_t^{\text{ch}} \leq u_t P_{\text{ch}}^{\text{max}}, \quad t \in \mathcal{T}, \quad (1c)$$

$$0 \leq p_t^{\text{dc}} \leq (1 - u_t) P_{\text{dc}}^{\text{max}}, \quad t \in \mathcal{T}, \quad (1d)$$

$$u_t \in \{0, 1\}, \quad t \in \mathcal{T}, \quad (1e)$$

$$\underline{e}_t \leq e_t \leq \bar{e}_t, \quad t \in \mathcal{T}, \quad (1f)$$

$$e_T = e_0. \quad (1g)$$

Here, C_t^{CH} and C_t^{DC} denote the charging and discharging cost coefficients, respectively; $\eta^{\text{ch}}, \eta^{\text{dc}} \in (0, 1]$ are the corresponding efficiencies; $P_{\text{ch}}^{\text{max}} > 0$ and $P_{\text{dc}}^{\text{max}} > 0$ are the charging and discharging power limits, respectively; $\Delta t > 0$ is the dispatch interval; and e_0 is the initial state of charge (SoC). For notational compactness, define $\alpha := P_{\text{ch}}^{\text{max}} / P_{\text{dc}}^{\text{max}}$.

Let Q^{R} denote the optimal value of the relaxed problem where (1e) is replaced with $0 \leq u_t \leq 1$. Equivalently, constraints (1c)–(1d) are recast to

$$p_t^{\text{ch}} \geq 0, \quad p_t^{\text{dc}} \geq 0, \quad p_t^{\text{ch}} + \alpha p_t^{\text{dc}} \leq P_{\text{ch}}^{\text{max}}. \quad (2)$$

Indeed, multiplying the upper bound in (1d) by α and adding it to (1c) yields the last inequality in (2). Conversely, if (2) holds, setting $u_t = p_t^{\text{ch}} / P_{\text{ch}}^{\text{max}} \in [0, 1]$ satisfies all constraints of the relaxed problem. Since the relaxed feasible set contains the original feasible set, $Q^{\text{R}} \leq Q$; the relaxation is called *optimal-value exact* if $Q^{\text{R}} = Q$. Define $\eta := \eta^{\text{ch}} \eta^{\text{dc}}$. We next characterize all relaxed charging/discharging actions producing a fixed SoC transition.

Complementary action. For a prescribed SoC transition, define $\Delta e_t := e_t - e_{t-1}$. The complementary charging and discharging powers that realize this transition are

$$p_t^{\text{ch},0} := [\Delta e_t]^+ / (\eta^{\text{ch}} \Delta t), \quad p_t^{\text{dc},0} := \eta^{\text{dc}} [-\Delta e_t]^+ / \Delta t, \quad (3)$$

respectively, where $[x]^+ := \max\{x, 0\}$.

Lemma 1 (Zero-SoC Cycling Decomposition): Fix a feasible SoC transition Δe_t . Any relaxed pair $(p_t^{\text{ch}}, p_t^{\text{dc}})$ satisfying (1b) for this transition and (2) can be uniquely represented as

$$p_t^{\text{ch}} = p_t^{\text{ch},0} + z_t, \quad (4)$$

$$p_t^{\text{dc}} = p_t^{\text{dc},0} + \eta z_t, \quad (5)$$

for a scalar z_t satisfying

$$0 \leq z_t \leq \bar{z}_t(\Delta e_t), \quad (6)$$

where

$$\bar{z}_t(\Delta e_t) := (P_{\text{ch}}^{\text{max}} - p_t^{\text{ch},0} - \alpha p_t^{\text{dc},0}) / (1 + \alpha\eta) \quad (7)$$

Proof: If $\Delta e_t \geq 0$, then $p_t^{\text{dc},0} = 0$ and $p_t^{\text{ch},0} = \Delta e_t / (\eta^{\text{ch}} \Delta t)$. Since $(p_t^{\text{ch}}, p_t^{\text{dc}})$ produces the same transition,

$$\Delta e_t = \eta^{\text{ch}} p_t^{\text{ch}} \Delta t - \frac{p_t^{\text{dc}} \Delta t}{\eta^{\text{dc}}} = \eta^{\text{ch}} p_t^{\text{ch},0} \Delta t.$$

Rearranging gives $p_t^{\text{dc}} = \eta (p_t^{\text{ch}} - p_t^{\text{ch},0})$. Thus $z_t := p_t^{\text{ch}} - p_t^{\text{ch},0} = p_t^{\text{dc}} / \eta \geq 0$, and (4)–(5) follow.

If $\Delta e_t < 0$, then $p_t^{\text{ch},0} = 0$. Equalities (1b) and (3) give $p_t^{\text{dc}} = p_t^{\text{dc},0} + \eta p_t^{\text{ch}}$. Defining $z_t := p_t^{\text{ch}} \geq 0$ yields (4)–(5).

The additional pair $(z_t, \eta z_t)$ produces zero SoC change since $\eta^{\text{ch}} z_t \Delta t - \eta z_t \Delta t / \eta^{\text{dc}} = 0$. Substituting (4)–(5) into (2) gives

$$p_t^{\text{ch},0} + \alpha p_t^{\text{dc},0} + (1 + \alpha\eta) z_t \leq P_{\text{ch}}^{\text{max}},$$

which is equivalent to (6)–(7). Uniqueness follows from $z_t = p_t^{\text{ch}} - p_t^{\text{ch},0} = (p_t^{\text{dc}} - p_t^{\text{dc},0}) / \eta$. ■

The scalar z_t measures the zero-SoC cycling amount, i.e., the simultaneous charging and discharging component that leaves the SoC transition unchanged. $\bar{z}_t(\Delta e_t)$ is the maximum cycling amount permitted by the remaining power headroom for the prescribed SoC transition. Moreover, $\bar{z}_t(\Delta e_t) = 0$ if and only if the transition uses the full charging or discharging power capacity, equivalently, $\Delta e_t = \eta^{\text{ch}} P_{\text{ch}}^{\text{max}} \Delta t$ or $\Delta e_t = -P_{\text{dc}}^{\text{max}} \Delta t / \eta^{\text{dc}}$; in this case, SCD is physically impossible for that transition.

III. NECESSARY AND SUFFICIENT CONDITIONS FOR RELAXATION EXACTNESS

This section first characterizes stagewise relaxation exactness for a fixed SoC transition, and then lift the result to the full horizon. For a fixed SoC transition Δe_t , define the complementary stage cost $c_t(\Delta e_t) := C_t^{\text{CH}} p_t^{\text{ch},0} + C_t^{\text{DC}} p_t^{\text{dc},0}$. Using (4)–(5),

$$C_t^{\text{CH}} p_t^{\text{ch}} + C_t^{\text{DC}} p_t^{\text{dc}} = c_t(\Delta e_t) + (C_t^{\text{CH}} + \eta C_t^{\text{DC}}) z_t.$$

Define the marginal cycling cost

$$\kappa_t^0 := C_t^{\text{CH}} + \eta C_t^{\text{DC}}. \quad (8)$$

For the fixed transition Δe_t , the original problem forces $z_t = 0$, while the relaxation chooses z_t over the interval (6). Hence the relaxed stage value is

$$\begin{aligned} c_t^{\text{R}}(\Delta e_t) &:= c_t(\Delta e_t) + \min_{0 \leq z_t \leq \bar{z}_t(\Delta e_t)} \kappa_t^0 z_t \\ &= c_t(\Delta e_t) - [-\kappa_t^0]^+ \bar{z}_t(\Delta e_t). \end{aligned} \quad (9)$$

Thus $c_t^{\text{R}}(\Delta e_t) = c_t(\Delta e_t)$ if and only if

$$\kappa_t^0 \geq 0 \quad \text{or} \quad \bar{z}_t(\Delta e_t) = 0. \quad (10)$$

We next lift this stagewise characterization to the full horizon using the feasible SoC trajectory set. The original and relaxed problems have the same projection onto the SoC trajectory space. Let

$$\begin{aligned} \mathcal{E} := \{e = (e_0, \dots, e_T) : & \underline{e}_t \leq e_t \leq \bar{e}_t, \quad t \in \mathcal{T}, \\ & -P_{\text{dc}}^{\text{max}} \Delta t / \eta^{\text{dc}} \leq e_t - e_{t-1} \leq \eta^{\text{ch}} P_{\text{ch}}^{\text{max}} \Delta t, \quad t \in \mathcal{T}, \\ & e_T = e_0\}, \end{aligned}$$

where the initial SoC e_0 is fixed. Indeed, every feasible solution of either model has an SoC trajectory in $\mathcal{E}$. Conversely, for any $e \in \mathcal{E}$, define the complementary powers by (3). The transition bounds in $\mathcal{E}$ imply $0 \leq p_t^{\text{ch},0} \leq P_{\text{ch}}^{\text{max}}$ and $0 \leq p_t^{\text{dc},0} \leq P_{\text{dc}}^{\text{max}}$. Moreover, $p_t^{\text{ch},0} p_t^{\text{dc},0} = 0$, so one can choose a binary mode u_t to satisfy (1c)–(1e). Hence e is implementable in the original problem.

Consequently, the two optimal values can be written as

$$\begin{aligned} Q &= \min_{e \in \mathcal{E}} \sum_{t \in \mathcal{T}} c_t(e_t - e_{t-1}), \\ Q^{\text{R}} &= \min_{e \in \mathcal{E}} \sum_{t \in \mathcal{T}} c_t^{\text{R}}(e_t - e_{t-1}). \end{aligned}$$

Theorem 1 (Necessary and Sufficient Exactness Condition): The relaxation is optimal-value exact, i.e., $Q^{\text{R}} = Q$, if and only if the relaxed problem admits an optimal SoC trajectory $e^* = (e_0^*, \dots, e_T^*) \in \mathcal{E}$ such that

$$[-\kappa_t^0]^+ \bar{z}_t(e_t^* - e_{t-1}^*) = 0, \quad \forall t \in \mathcal{T}. \quad (11)$$

Equivalently, for every $t \in \mathcal{T}$, $\kappa_t^0 \geq 0$ or $\bar{z}_t(e_t^* - e_{t-1}^*) = 0$.

Proof: 1) *Sufficiency:* Suppose that the relaxed problem has an optimal SoC trajectory e^* satisfying (11). By (9), $c_t^{\text{R}}(e_t^* - e_{t-1}^*) = c_t(e_t^* - e_{t-1}^*)$ for all $t \in \mathcal{T}$. Thus

$$Q \leq \sum_{t \in \mathcal{T}} c_t(e_t^* - e_{t-1}^*) = \sum_{t \in \mathcal{T}} c_t^{\text{R}}(e_t^* - e_{t-1}^*) = Q^{\text{R}}.$$

Together with $Q^{\text{R}} \leq Q$, this gives $Q^{\text{R}} = Q$.

2) *Necessity:* Suppose $Q^{\text{R}} = Q$, and let $e^{\text{O}} \in \mathcal{E}$ be an optimal SoC trajectory of the original problem. Then $\sum_{t \in \mathcal{T}} c_t(e_t^{\text{O}} - e_{t-1}^{\text{O}}) = Q$. The same trajectory is feasible for the relaxed projection, and (9) implies $c_t^{\text{R}}(\Delta e_t) \leq c_t(\Delta e_t)$ for every feasible transition. Hence

$$Q^{\text{R}} \leq \sum_{t \in \mathcal{T}} c_t^{\text{R}}(e_t^{\text{O}} - e_{t-1}^{\text{O}}) \leq \sum_{t \in \mathcal{T}} c_t(e_t^{\text{O}} - e_{t-1}^{\text{O}}) = Q.$$

Since $Q^{\text{R}} = Q$, both inequalities in the preceding display are equalities. Therefore e^{O} is also optimal for the relaxed problem and $\sum_{t \in \mathcal{T}} [c_t(e_t^{\text{O}} - e_{t-1}^{\text{O}}) - c_t^{\text{R}}(e_t^{\text{O}} - e_{t-1}^{\text{O}})] = 0$. Each summand is nonnegative, because by (9),

$$c_t(e_t^{\text{O}} - e_{t-1}^{\text{O}}) - c_t^{\text{R}}(e_t^{\text{O}} - e_{t-1}^{\text{O}}) = [-\kappa_t^0]^+ \bar{z}_t(e_t^{\text{O}} - e_{t-1}^{\text{O}}) \geq 0.$$

Hence every summand is zero; namely, $[-\kappa_t^0]^+ \bar{z}_t(e_t^{\text{O}} - e_{t-1}^{\text{O}}) = 0$ for all $t \in \mathcal{T}$. Thus, the relaxed problem admits the optimal trajectory e^{O} satisfying (11). ■

IV. PHYSICAL INTERPRETATION AND ALGORITHM

This section interprets the necessary and sufficient exactness condition and proposes a two-stage relaxation-and-recovery algorithm for solving Problem (1).

For an optimal relaxed SoC trajectory $e^* \in \mathcal{E}$, the optimal value of the relaxed full-horizon problem is expressed by

$$Q^{\text{R}} = \min_{e \in \mathcal{E}} \sum_{t=1}^T \left[c_t(\Delta e_t) + \min_{0 \leq z_t \leq \bar{z}_t(\Delta e_t)} \kappa_t^0 z_t \right]. \quad (12)$$

Hence the only possible gap from the original model is the zero-SoC cycling term. Here, κ_t^0 is the marginal cycling cost,

TABLE I
STAGewise BEHAVIOR OF THE RELAXED ESS ACTION

κ_t^0	$\bar{z}_t$	Relaxed optimum z_t^*	SCD possible?	Exact for this SoC transition?
> 0	Any	$z_t^* = 0$	No	Yes
$= 0$	> 0	Any $z_t^* \in [0, \bar{z}_t]$	Yes, possible	Yes, value exact
$= 0$	$= 0$	$z_t^* = 0$	No	Yes
< 0	$= 0$	$z_t^* = 0$	No	Yes
< 0	> 0	$z_t^* = \bar{z}_t > 0$	Yes	No

while $\bar{z}_t(\Delta e_t^*)$ is the available physical headroom for such cycling. By Theorem 1, exactness along e^* is equivalent to

$$\kappa_t^0 \geq 0 \quad \text{or} \quad \bar{z}_t(\Delta e_t^*) = 0, \quad \forall t \in \mathcal{T}. \quad (13)$$

Table I summarizes the possible cases. When $\kappa_t^0 > 0$, cycling is strictly penalized and the relaxed optimum satisfies $z_t^* = 0$. When $\kappa_t^0 = 0$ and $\bar{z}_t > 0$, an SCD solution may occur since all $z_t \in [0, \bar{z}_t]$ have the same cost. The SCD component can be removed by setting $z_t = 0$ in (4)–(5), so the returned SCD solution is only a degenerate representation of an exact optimum.

If $\kappa_t^0 < 0$, the relaxed stage problem favors the largest feasible cycling amount, $z_t^* = \bar{z}_t(\Delta e_t^*)$. If $\bar{z}_t(\Delta e_t^*) = 0$, the selected transition already uses the full charging or discharging power capacity, and the relaxation remains exact. Otherwise, $z_t^* > 0$ and the relaxed solution exploits profitable SCD. After solving the relaxed problem, $z_t^* = \min\{p_t^{\text{ch},*}, p_t^{\text{dc},*}/\eta\}$; a positive value indicates that binary enforcement is required.

These observations motivate Algorithm 1. The fully relaxed problem (1) is first solved to compute the cycling amount z_t at each period. If $\kappa_t^0 > 0$, SCD is excluded in any optimum; if $\kappa_t^0 = 0$, SCD does not change the objective and can be removed by setting $z_t = 0$ in (4)–(5). Therefore, if no period in $\mathcal{T}^- := \{t \in \mathcal{T} : \kappa_t^0 < 0\}$ exhibits cycling, the relaxed solution can be recovered directly. Otherwise, the problem is resolved as an MIP with binary modes imposed only on $\mathcal{T}^-$, followed by the same recovery at zero-cost periods. Thus, at most one LP and one partial MIP problem need to be solved, and the resulting schedule is complementary and optimal.

Algorithm 1 Two-Stage Relaxation-and-Recovery Method

Input: ESS parameters

Output: Optimal complementary schedule $\{e_t^*, p_t^{\text{ch},*}, p_t^{\text{dc},*}\}_{t \in \mathcal{T}}$

- 1: Compute $\eta = \eta^{\text{ch}}\eta^{\text{dc}}$ and $\kappa_t^0 = C_t^{\text{CH}} + \eta C_t^{\text{DC}}$.
- 2: Set $\mathcal{T}^- = \{t \in \mathcal{T} : \kappa_t^0 < 0\}$ and $\mathcal{T}^0 = \{t \in \mathcal{T} : \kappa_t^0 = 0\}$.
- 3: Solve the fully relaxed problem (1) to obtain $\{e_t, p_t^{\text{ch}}, p_t^{\text{dc}}\}_{t \in \mathcal{T}}$.
- 4: Compute $z_t = \min\{p_t^{\text{ch}}, p_t^{\text{dc}}/\eta\}$, $\forall t \in \mathcal{T}$, and set $\mathcal{V} = \{t \in \mathcal{T}^- : z_t > 0\}$.
- 5: **if** $\mathcal{V} \neq \emptyset$ **then**
- 6: Re-solve (1) with binary u_t on $\mathcal{T}^-$ and continuous u_t otherwise to update $\{e_t, p_t^{\text{ch}}, p_t^{\text{dc}}\}_{t \in \mathcal{T}}$.
- 7: Recompute z_t , $\forall t \in \mathcal{T}^0$.
- 8: **end if**
- 9: **for** $t \in \mathcal{T}^0$ with $z_t > 0$ **do**
- 10: Recover the complementary powers by (3) with $\Delta e_t = e_t - e_{t-1}$ (equivalently, set $z_t = 0$ in (4)–(5)).
- 11: **end for**
- 12: **return** The optimal schedule $\{e_t^*, p_t^{\text{ch},*}, p_t^{\text{dc},*}\}_{t \in \mathcal{T}}$

V. NUMERICAL EXPERIMENTS

We consider a five-period ESS power dispatch with parameters $\Delta t = 1$, $\eta^{\text{ch}} = \eta^{\text{dc}} = 0.95$, $P_{\text{ch}}^{\text{max}} = P_{\text{dc}}^{\text{max}} = 6$, $\underline{e}_t = 0$,

TABLE II
STEP 1: RELAXED LP REPRESENTATIVE SOLUTION

t	C_t^{CH}	C_t^{DC}	κ_t^0	p_t^{ch}	p_t^{dc}	e_t	z_t	SCD
1	10	-5	5.4875	0	3.5000	6.3158	0	No
2	10	-5	5.4875	0	6	0	0	No
3	2.5000	-5	-2.0125	3.1537	2.8463	0	3.1537	Yes
4	1	5	5.5125	6	0	5.7000	0	No
5	2.7075	-3	0	5.3009	0.6991	10	0.7746	Yes

* Units: costs and κ_t^0 in \$/MWh; powers and z_t in MW; e_t in MWh.

TABLE III
STEP 2: PARTIAL MIP SOLUTION WITH BINARY MODE AT $t = 3$

t	C_t^{CH}	C_t^{DC}	κ_t^0	u_t	p_t^{ch}	p_t^{dc}	e_t	z_t	SCD
1	10	-5	5.4875	–	0	4.7500	5	0	No
2	10	-5	5.4875	–	0	4.7500	0	0	No
3	2.5000	-5	-2.0125	1	6	0	5.7000	0	No
4	1	5	5.5125	–	6	0	11.4000	0	No
5	2.7075	-3	0	–	2.4547	3.5453	10	2.4547	Yes

TABLE IV
STEP 3: RECOVERED COMPLEMENTARY SOLUTION

t	C_t^{CH}	C_t^{DC}	κ_t^0	p_t^{ch}	p_t^{dc}	e_t	z_t	SCD
1	10	-5	5.4875	0	4.7500	5	0	No
2	10	-5	5.4875	0	4.7500	0	0	No
3	2.5000	-5	-2.0125	6	0	5.7000	0	No
4	1	5	5.5125	6	0	11.4000	0	No
5	2.7075	-3	0	0	1.3300	10	0	No

$\bar{e}_t = 20$, and $e_0 = e_T = 10$. The cost vectors are $C^{\text{CH}} = [10, 10, 2.5, 1, 2.7075]$ and $C^{\text{DC}} = [-5, -5, -5, 5, -3]$, which give $\kappa^0 = [5.4875, 5.4875, -2.0125, 5.5125, 0]$. All LP and MIP models are solved by Gurobi. Algorithm 1 is applied as follows. In Step 1, the relaxed LP returns SCD at periods $t = 3$ and $t = 5$, as shown in Table II. Consistent with the conclusion in Table I, since $\kappa_3^0 < 0$ and $z_3 > 0$, the relaxation exploits profitable SCD that attains the lower objective value -35.5919 . In contrast, when $\kappa_5^0 = 0$, the SCD at $t = 5$ does not change the objective value, as indicated by (9). In Step 2, we impose a binary mode only at $t = 3$ and keep the other periods relaxed. The resulting partial MIP, reported in Table III, has objective -30.4900 , matching the full MIP objective obtained by enforcing binary modes in all periods. Although SCD remains at $t = 5$, it can be removed by (3). Table IV shows the recovered complementary solution, whose objective remains -30.4900 . The experiment verifies the correctness of the proposed theory and algorithm.

VI. CONCLUSION

This letter provides a necessary and sufficient condition for the relaxation exactness of a complementarity-constrained ESS power dispatch problem. It shows that the relaxation gap is governed by the marginal cycling cost κ_t^0 and the available cycling headroom $\bar{z}_t$. In particular, the stagewise gap is $[-\kappa_t^0]^+ \bar{z}_t$, so SCD creates a true gap only when it is both profitable and physically feasible. Therefore, the presence of SCD in a relaxed solution does not necessarily mean that the relaxation is inexact. When $\kappa_t^0 = 0$, SCD is only a degenerate zero-cost cycling component, and setting $z_t = 0$ recovers a complementary optimal schedule with the same SoC trajectory and objective value. The proposed condition can guide solver design by identifying unnecessary ESS mode binaries and

enabling recovery of feasible solutions, thereby improving computational efficiency. Future work will integrate this theory into network-constrained power system optimization problems.

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